

GROUP PROJECT GUIDELINE

TOPIC 3: INVENTORY MANAGEMENT WITH AI-POWERED DEMAND FORECASTING

OVERVIEW

Course: TMA Advanced AI Training

Type: Group Projects (10-15 members per group)

Objectives: This project aims at giving learners experience on building an end-to-end AI solution for real-world applications through the use of public datasets. Students will work with publicly available inventory and sales data to practice the full development cycle—starting from data preprocessing and exploratory analysis, to designing and training time-series forecasting models, evaluating model performance, and deploying the solution via an interactive dashboard. The ultimate goal is to develop an AI-powered demand forecasting system that helps manufacturing businesses optimize inventory levels, reduce stockouts and overstock, and support smarter restocking decisions through accurate and actionable predictions.

DATASET

Learners may either **use a dataset of their choice** or select from one of the recommended datasets listed below:

No.	Name	Source	Description
1	Store Sales - Time Series Forecastint	Link	Retail sales data from Ecuador, includes holidays, oil prices, promotions.
2	Rossmann Store Sales	Link	Sales from 1,000+ stores with store, promotion, and holiday info.
3	M5 Forecasting - Accuracy	Link	Walmart sales data across multiple products and regions.
4	Retail Store Inventory Forecasting Dataset	Link	This dataset contains over 73,000 rows of daily data across multiple stores and products.
5	Manufacturing sales, orders, inventory dataset	Link	Canadian Sales of goods manufactured (shipments), new orders, unfilled orders, inventories, raw materials, goods or work in process, finished goods.

GUIDELINES

Each group will submit 1 report with the following constraints:

- Report must include reference/citation of the sources being used
- Visualization can be included in the report but must not exceed 50% of the total space
- Report can range from 10 to 15 pages

REPORT GUIDELINE

Your final report should be structured according to the **CRISP-DM (Cross-Industry Standard Process for Data Mining)** standard ([Link](#)) and should include the following sections:

- 1. Business Understanding:**
 - Clearly describe the business use case.
 - Explain the challenges in inventory management and how AI/ML can help address them.
 - Define the project objectives and success criteria from a business perspective.
- 2. Data Understanding:**
 - Describe the dataset(s) used, including structure, types of data, and sources.
 - Explore patterns in the data: trends, seasonality, anomalies, etc.
 - Identify and describe external factors (e.g., holidays, promotions) that may influence demand.
- 3. Data Preparation:**
 - Explain the preprocessing steps: handling missing values, outliers, and data cleaning.
 - Describe how you formatted timestamps (daily, weekly, etc.).
 - Highlight key feature engineering steps (e.g., lag variables, rolling averages, holiday flags).
 - Summarize insights gained during data preparation.
- 4. Modeling:**
 - Describe the AI/ML techniques used (e.g., LSTM, Prophet, ARIMA).
 - Explain how the data was split and how the model was trained.
 - Provide details on hyperparameter tuning and model configuration.
- 5. Evaluation:**
 - Present the performance metrics (e.g., RMSE, MAPE) achieved by your models.
 - Compare results across different models or configurations.
 - Analyze the strengths and weaknesses of your final model(s).
- 6. Deployment/Delivery:**
 - Describe the final output: dashboard or visual tool.
 - Share key findings and insights from the project.
 - Discuss how your solution can be applied in real-world scenarios.
 - Suggest future improvements or next steps.

GRADING GUIDELINE

No.	Name	Description	Grading Percentage
1	Data	Clearly defined data sources and collection methods	30%
2	AI/ML Technology	Explanation of AI model, techniques and implementation	40%
3	Demo	Functional code demonstrating the AI solution	20%
4	Reference & Citation	At least 5 credible sources supporting the project	10%